

# Risks related to AI & AI and Cybersecurity



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- Harms possible from AI
- AI Lifecycle
- Benefits of AI in Cybersecurity
- AI Use cases in Cybersecurity
- State of the World Adapting Gen-AI
- Enterprise Best Practices for GenAI

# Labs

- Students to use Gemini for different prompts
- Asking the AI Models to create a phishing email

# Harms possible from AI

- AI systems can deliver significant value, but when misused, poorly designed, or insufficiently governed, they can cause technical, social, ethical, and economic harm.
- Like:
  - ✓ Misinformation and Disinformation
  - ✓ Bias and Discrimination
  - ✓ Privacy and Data Leakage
  - ✓ Security Threats
  - ✓ Environmental Impact
  - ✓ Hallucinations and Incorrect Outputs
  - ✓ Loss of Human Control
  - ✓ Job Displacement and Economic Impact
  - ✓ Ethical and Moral Concerns
  - ✓ Malicious Use and Weaponization

AI is powerful but not neutral—its impact depends on how responsibly it is designed, deployed, and governed.



# Harms possible - AI-Enabled Cybercrime

- **Automated phishing & social engineering**
  - Highly personalized spear-phishing using OSINT + LLMs
  - Deepfake voice/video scams (CEO fraud, KYC bypass)
- **Malware development**
  - Polymorphic malware generation
  - AI-assisted exploit development and obfuscation
- **Credential attacks**
  - Intelligent password spraying
  - CAPTCHA bypass using vision models

Impact: Increased attack scale, speed, and success rate.

# Harms possible - Adversarial Attacks on AI Systems

- AI systems themselves can be attacked.
  - Prompt injection
  - Model poisoning
  - Data poisoning
  - Evasion attacks (adversarial examples)
  - Model extraction & IP theft
- Impact: Incorrect decisions, security control bypass, loss of trust.



# Harms possible - Over-Automation & Blind Trust

- Security teams relying blindly on AI alerts
  - AI hallucinations leading to incorrect decisions
  - Missed threats due to model bias or training gaps
- Impact: False confidence, delayed incident response.

# Harms possible - Mass Data Harvesting

- Training on scraped personal data
  - Leakage of sensitive information via model outputs
  - Memorization of PII in large models
- **Impact:** GDPR, DPDP Act (India), HIPAA, PCI violations.



# Harms possible - Surveillance & Tracking

- Facial recognition misuse
  - AI-powered behavioral monitoring
  - Predictive policing risks
- **Impact:** Erosion of civil liberties and privacy.

# Harms possible - Misinformation & Deepfakes

- Fake news generation at scale
  - Political manipulation
  - Fake audio/video evidence
- **Impact:** Democratic instability, public mistrust.

# Harms possible - Manipulation & Persuasion

- AI-generated propaganda
- Targeted psychological influence
- Echo chambers amplified by algorithms
- **Impact:** Radicalization, social polarization.

# Harms possible - Market Manipulation

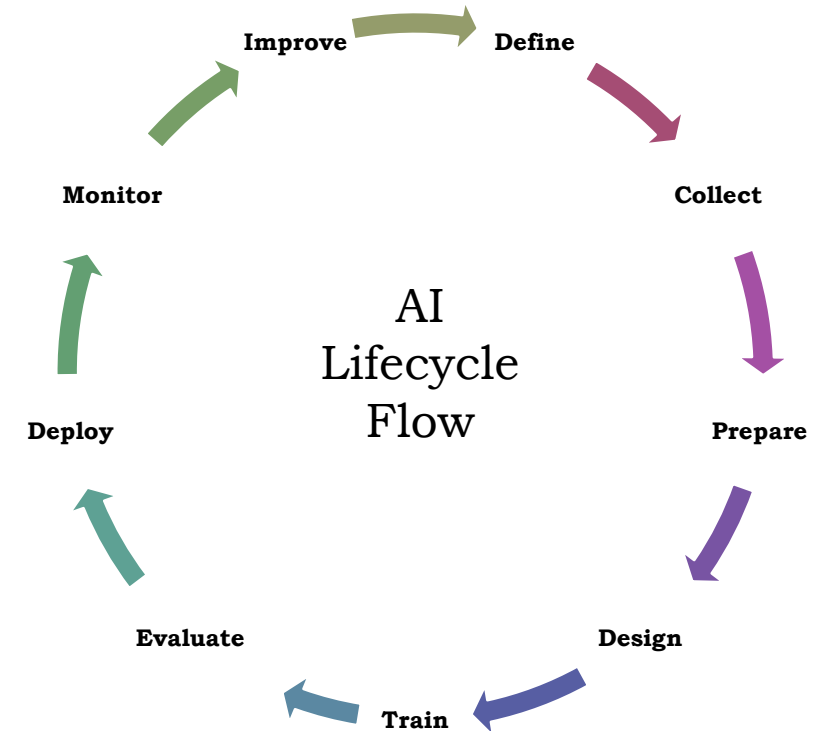
- AI-driven trading bots
- Fake reviews and rating manipulation
- Price manipulation via automated systems
- **Impact:** Financial instability, consumer deception.



# AI Lifecycle

- The AI lifecycle defines the **end-to-end process** of building, deploying, operating, and governing AI systems in a controlled and responsible manner.

- ✓ Problem Definition & Business Understanding
- ✓ Data Collection
- ✓ Data Preparation (Data Engineering)
- ✓ Model Selection & Design
- ✓ Model Training
- ✓ Model Evaluation & Validation
- ✓ Deployment
- ✓ Monitoring & Performance Management
- ✓ Feedback & Continuous Improvement
- ✓ Governance, Risk & Compliance (Across All Stages)



# Benefits of AI in Cybersecurity

Faster Threat  
Detection

Advanced  
Malware  
Detection

Phishing and  
Social  
Engineering  
Prevention

Automated  
Incident  
Response

Behavioral  
Analytics

Security  
Operations  
Center (SOC)  
Efficiency

Vulnerability  
Management

Cloud and  
Network  
Security

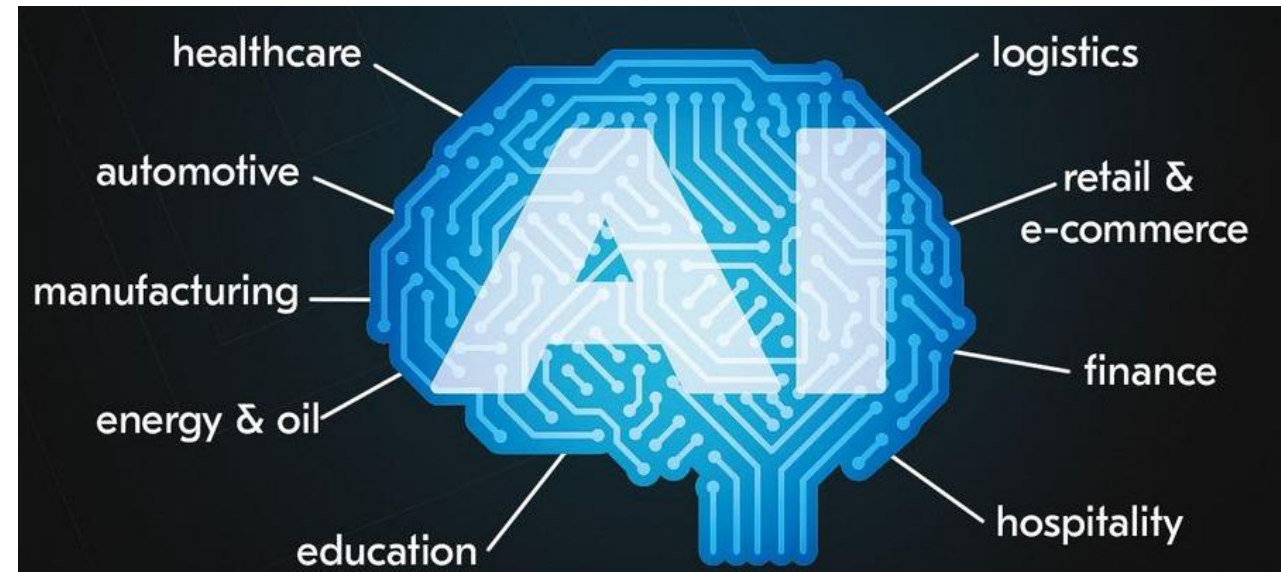
Threat  
Intelligence  
Enhancement

Scalability and  
24×7 Protection



# AI Use cases in Cybersecurity

- Threat Detection & Anomaly Detection
- Malware Detection
- Phishing & Email Security
- User and Entity Behavior Analytics (UEBA)
- Security Operations Center (SOC) Automation
- Automated Incident Response (SOAR)
- Vulnerability Management
- Identity & Access Management (IAM)
- Network Security & Traffic Analysis
- Cloud Security



# AI Use cases in Cybersecurity - Security Operations Center (SOC)

## ▪ **Threat Detection & Alert Correlation**

- AI analyzes large volumes of logs and telemetry to identify malicious patterns.
- Use Cases
  - Detect anomalous behavior across endpoints, network, and cloud
  - Correlate alerts from SIEM, EDR, NDR, and identity systems

## ▪ **Incident Prioritization & Triage**

- AI helps SOC analysts focus on high-risk incidents.
- Use Cases
  - Risk-based alert scoring
  - Automated incident classification (malware, phishing, insider threat)



# AI Use cases in Cybersecurity - Endpoint Security (EDR/XDR)

## ▪ **Behavioral Malware Detection**

- AI detects malware based on behavior rather than signatures.
- Use Cases
  - Detect zero-day malware
  - Identify ransomware encryption patterns

## ▪ **Automated Response**

- AI triggers containment actions.
- Use Cases
  - Isolate compromised endpoints
  - Kill malicious processes

# AI Use cases in Cybersecurity - Network Security

- **Network Traffic Analysis (NTA)**

- AI inspects network behavior rather than packet signatures.
- Use Cases
  - Detect C2 beaconing
  - Lateral movement detection
  - DNS tunneling detection

- **Intrusion Detection & Prevention**

- AI-enhanced IDS/IPS systems adapt to evolving threats.
- Use Cases
  - Identify zero-day exploits
  - Detect encrypted traffic anomalies



# AI Use cases in Cybersecurity - IAM

- **Adaptive Authentication**

- AI evaluates risk before granting access.

- Use Cases

- Detect credential theft
    - Enforce step-up authentication
    - Identify insider threats

- Examples

- Conditional Access policies

- Benefit: Zero Trust enforcement

# AI Use cases in Cybersecurity - Phishing & Email Security

## ▪ **Phishing Detection**

- AI analyzes email content, sender behavior, and intent.
- Use Cases
  - Detect spear-phishing and BEC attacks
  - Identify malicious URLs and attachments
  - Language and tone analysis

## ▪ **Deepfake Detection**

- AI identifies manipulated audio/video used in fraud.
- Use Cases
  - Voice-based CEO fraud detection
  - Video KYC verification



# AI Use cases in Cybersecurity - Threat Intelligence & Hunting

- **Automated Threat Intelligence Analysis**

- AI enriches and correlates threat feeds.
- Use Cases
  - IOC prioritization
  - Campaign attribution
  - TTP clustering

- **Proactive Threat Hunting**

- AI suggests hypotheses for analysts.
- Use Cases
  - Identify low-and-slow attacks
  - Detect persistence mechanisms



# AI Use cases in Cybersecurity - Vulnerability Management

- **Risk-Based Vulnerability Prioritization**

- AI scores vulnerabilities based on exploit likelihood.
- Use Cases
  - Combine CVSS, exploit availability, asset value
  - Predict real-world exploitation

- **Attack Surface Management**

- AI identifies exposed assets.
- Use Cases
  - Shadow IT discovery
  - Misconfigured cloud resources



# AI Use cases in Cybersecurity - Digital Forensics & IR

- **Log & Evidence Analysis**

- AI accelerates investigations.
- Use Cases
  - Timeline reconstruction
  - Root cause analysis
  - Evidence correlation

- **Post-Incident Learning**

- AI improves detection rules post-incident.

# State of the World Adapting Gen-AI

- **Global Adoption Status** - GenAI is becoming a general-purpose technology, similar to the internet or cloud.
- **Industry-Wise Adoption** - High adoption by technology & software. Healthcare, manufacturing
- **Enterprise Use Patterns** - GenAI is embedded into daily workflows, not used as a standalone tool.
- **Workforce Impact** - Redefinition of job roles rather than mass replacement
- **Regional Perspective** - North America, Europe, Asia-Pacific, India
- **Governance, Regulation & Trust** - Balancing innovation with risk management.
- **Security & Risk Awareness** - Adoption of human-in-the-loop and monitoring controls
- **Technology Maturity** - Integration with cloud, data platforms, and enterprise tools





# Enterprise Best Practices for GenAI

- Start with Clear Business Use Cases
- Establish Strong Data Governance
- Choose the Right GenAI Model Strategy
- Implement Responsible AI Principles
- Keep Humans in the Loop
- Prepare the Workforce
- Secure the GenAI Stack
- Manage Hallucinations and Accuracy
- Integrate with Existing Enterprise Systems
- Monitor, Measure, and Improve Continuously
- Ensure Regulatory and Legal Compliance







*That's all Folks!*